

AntiBanana

Deepfake Detection & Protection System

Full Project Report

2025-11-22

Executive Summary

AntiBanana is a cutting-edge deepfake detection and protection system designed to counter the growing threat of AI-generated media. It combines advanced detection algorithms with a novel protection framework known as MMHI (Multi-Modal Hallucination Injection). The system is built with a modern Next.js frontend and a robust Python/Flask backend, leveraging state-of-the-art machine learning models to identify and neutralize deepfake threats.

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Chapter 1 : System Architecture

The AntiBanana system follows a modern client-server architecture:

1. Frontend (Next.js):

- A responsive, high-performance web interface built with React and TypeScript.
- Features real-time analysis dashboards, drag-and-drop file uploads, and interactive visualizations.
- Integrates specialized components for displaying detection results, including heatmaps and confidence scores.

2. Backend (Python/Flask):

- A scalable API server handling image processing and model inference.
- Integrates PyTorch for deep learning operations.
- Modular design allows for easy addition of new detection methods and protection phases.

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Chapter 2 : Deepfake Detection Capabilities

AntiBanana employs a multi-layered approach to detection:

1. Ensemble Detection Model:

- Utilizes a RexNet-150 architecture trained on 64x64 image patches for high efficiency and accuracy.
- Capable of detecting subtle artifacts introduced by GANs and Diffusion models.

2. SynthID Detection:

- Specialized module to detect Google's invisible SynthID watermarks.
- Analyzes the frequency domain to identify imperceptible signatures in AI-generated content.
- Provides confidence metrics and watermark strength indicators.

3. Explainable AI (XAI):

- Integrates GradCAM++, LIME, RISE, SHAP, and SOBOL methods.
- Generates heatmaps to visually explain WHICH parts of an image triggered the detection.
- Helps build user trust by providing transparency into the model's decision-making.

4. C2PA Provenance Verification:

- Checks for Content Credentials (C2PA) standards.
- Verifies the digital history and origin of media files.

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Chapter 3 : Deepfake Protection (MMHI)

The core innovation of AntiBanana is the Multi-Modal Hallucination Injection (MMHI) framework. This proactive defense mechanism "poisons" images to prevent them from being used to train deepfakes or to disrupt deepfake generation.

Key Features:

- 15-Phase Protection Pipeline: A comprehensive sequence of image transformations including Semantic Decoupling, Frequency Disruption, and Adversarial Noise.
- NanoBanana Breaker: A specialized module targeting transformer-based generators. It uses techniques like Feature Basin Trap Poisoning to cause catastrophic failure in generation models.
- Safety Trigger Injection: Embeds invisible text triggers to activate safety filters in AI models, preventing them from processing the protected image.

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Chapter 4 : Recent Technical Improvements

Recent development sprints have focused on stability and accuracy:

1. Model Architecture Fixes:

- Resolved mismatches between training (RexNet-150) and inference architectures.
- Implemented automatic architecture detection from checkpoint keys.

2. Preprocessing Alignment:

- Standardized image input size to 64x64 to match training data, significantly improving detection accuracy.
- Updated normalization pipelines for consistency.

3. Enhanced Visualization:

- Fixed XAI heatmap rendering to correctly overlay saliency maps on original images.
- Added dedicated UI components for SynthID detection results.

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Conclusion

AntiBanana represents a significant step forward in media integrity. By combining robust detection with aggressive protection mechanisms, it offers a dual-layer defense against the proliferation of deepfakes. The system is now fully operational with verified detection pipelines and a stable protection framework.